

NTRES 6100

Collaborative and Reproducible Data Science in R

Fall 2026

Today

Why are we here?

Why reproducible data science matters

Who are we?

Meet the teaching team and each other

What are we going to do?

Course goals, structure, and expectations

Then we will get situated with **R and RStudio**.

Is science facing a
reproducibility crisis?

nature International weekly journal of science

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1,500 scientists lift the lid on reproducibility

Survey sheds light on the 'crisis' rocking research.

[Monya Baker](#)

25 May 2016 | Corrected: 28 July 2016

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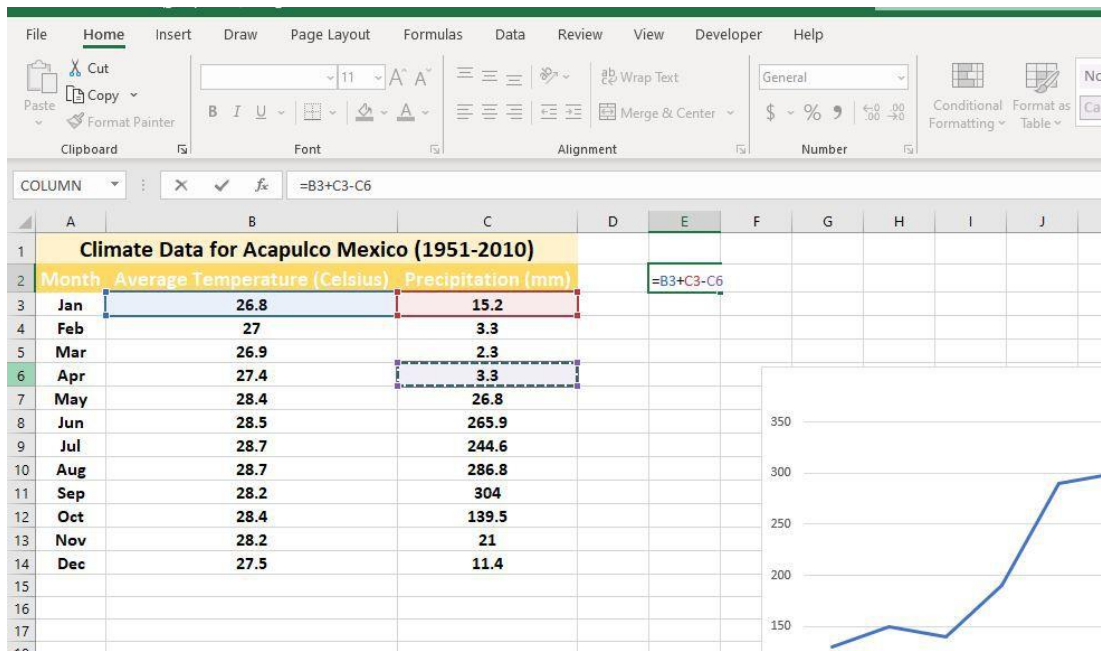
“More than 70% of researchers have tried and failed to reproduce another scientist's experiments, and more than half have failed to reproduce their own experiments”

Nature's survey of 1,576 researchers

Discuss in small groups

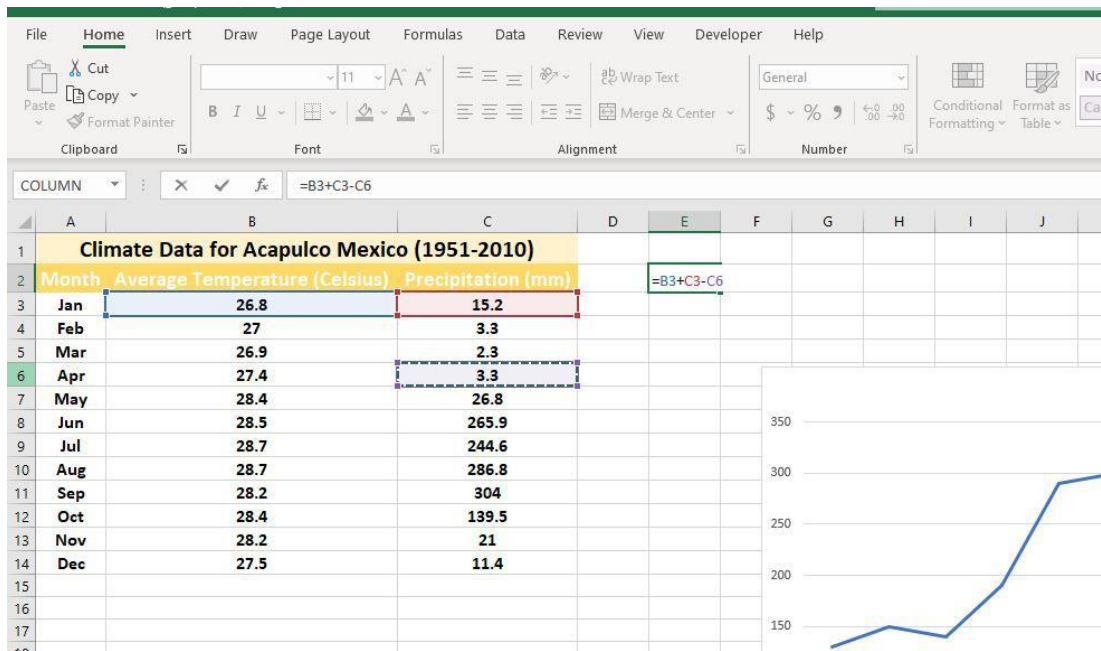
- Why may we sometimes fail to reproduce results in re-analysis of existing datasets?
- What can we do to ensure better reproducibility?

Point-and-click analysis leaves no paper trail



Point-and-click analysis leaves no paper trail

And can be tedious and error prone if repeated many times



Does this look familiar?

☐ Name	Date modified	Type
 Rscript_4_21_2016.R	5/1/2016 3:03 PM	R File
 Rscript_4_22_2016a.R	5/1/2016 3:03 PM	R File
 Rscript_4_22_2016b.R	5/1/2016 3:03 PM	R File
 Rscript_4_24_2016.R	5/1/2016 3:03 PM	R File
 Rscript_final.R	5/1/2016 3:03 PM	R File
 Rscript_final_final.R	5/1/2016 3:03 PM	R File
 Rscript_really_final.R	5/1/2016 3:03 PM	R File
 Rscript_really_really_final_final.R	5/1/2016 3:03 PM	R File

Reproducible workflows create a paper trail

Code

Records the analytical steps

Version control

Records how files change over time

Reproducible documents

Connect code, results, figures, and narrative

Tools for

better science in less time

Tools for

better science in less time

and with less pain

Who benefits?



Future you

Less re-learning, fewer mysteries, faster progress



Collaborators

Clearer workflows and easier handoffs



Science

More transparency and more opportunities to catch errors

Who are we?

Nina Overgaard Therkildsen
(Instructor)



Azwad Iqbal
(TA)



Who are you?

- Round of introductions
 - Your name
 - Your program
 - 2-3 words that best describe the type of data you work with (or expect to work with)

Tools for

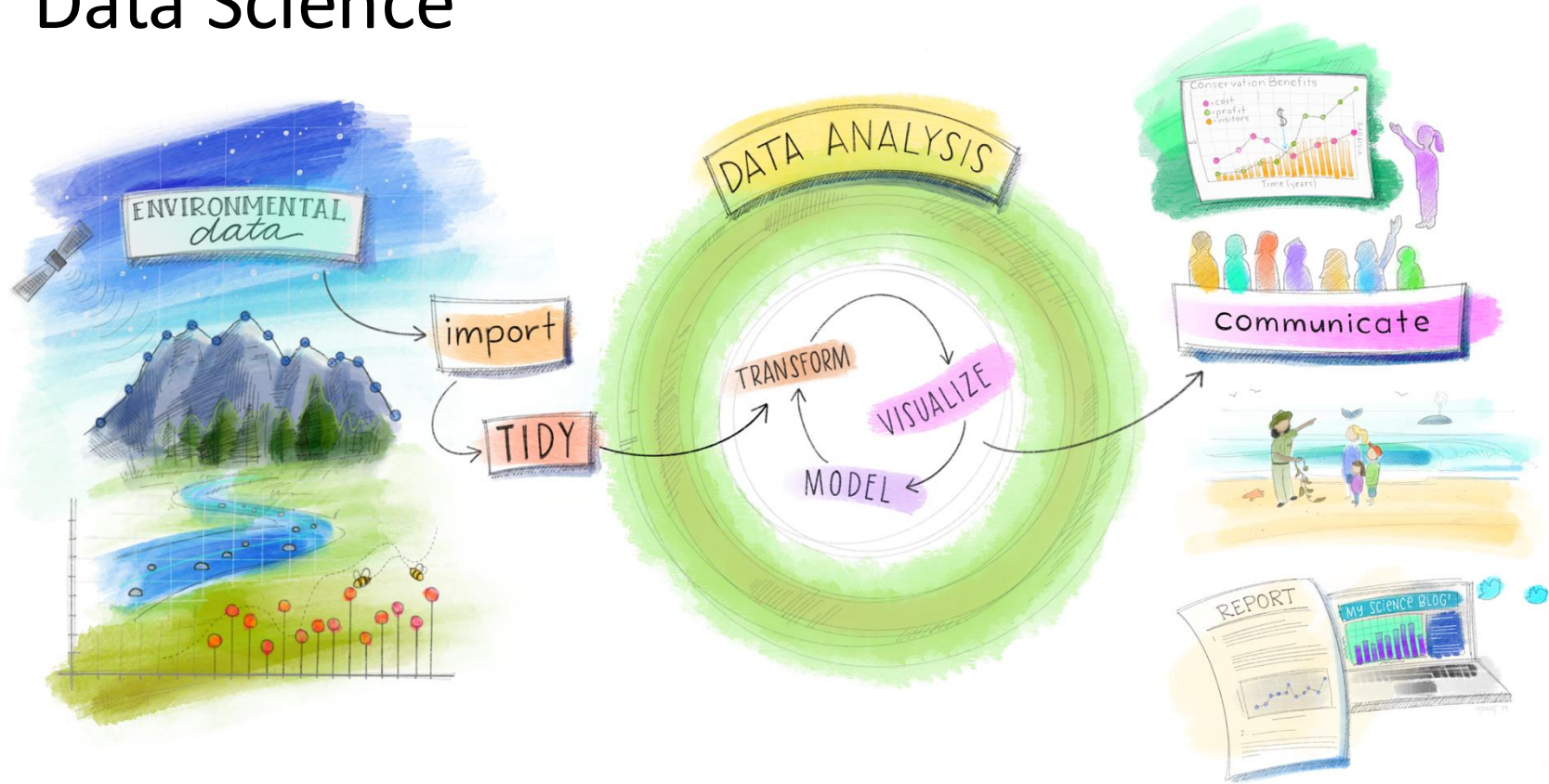
better science in less time

and with less pain

Data Science

Turning raw data into understanding, insight, and knowledge

Data Science



By the end of the course, you should be able to...

- build reproducible, project-oriented workflows in RStudio
- import, clean, reorganize, and combine messy data
- explore and visualize data effectively with **ggplot2**
- write simple functions and automate repeated tasks
- use **Git and GitHub** for version control and collaboration
- use **Quarto** to combine narrative, code, figures, and results
- use **AI as a coding partner while critically evaluating its output**

What this course is *not*

We will not focus on

- statistical inference
- modeling and simulation
- very large-scale computing
- domain-specific analysis pipelines

Data Science





For Big-Data Scientists, 'Janitor Work' Is Key Hurdle to Insights

By STEVE LOHR AUG. 17, 2014



Monica Rogati, Jawbone's vice president for data science, with Brian Wilt, a senior data scientist.
Peter DaSilva for The New York Times

Data scientists spend 50 - 80% of their time mired in this more mundane labor of collecting and preparing unruly digital data, before it can be explored for useful nuggets.

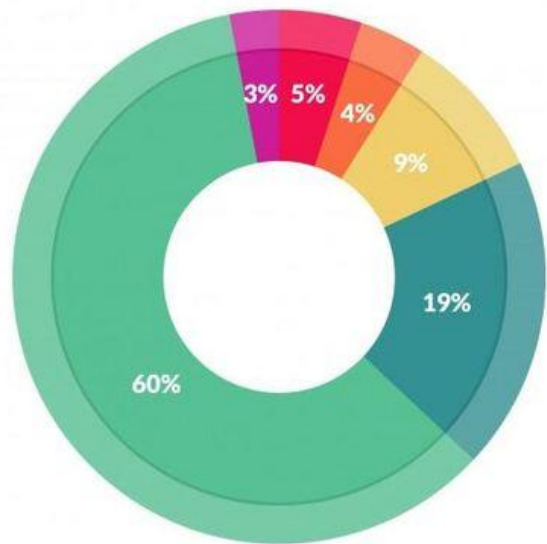


EMAIL

Technology revolutions come in measured, sometimes foot-dragging steps.
The lab science and marketing enthusiasm tend to underestimate the

Survey of 80 data scientists

<https://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/#25167ec06f63>



What data scientists spend the most time doing

- Building training sets: 3%
- Cleaning and organizing data: 60%
- Collecting data sets; 19%
- Mining data for patterns: 9%
- Refining algorithms: 4%
- Other: 5%

A major theme of this course

Do not build your analysis around whatever strange format your data happened to arrive in.

Instead, deliberately create structures that make analysis:

- easier to reason about
- easier to reuse
- easier to communicate
- easier to reproduce

Why R?

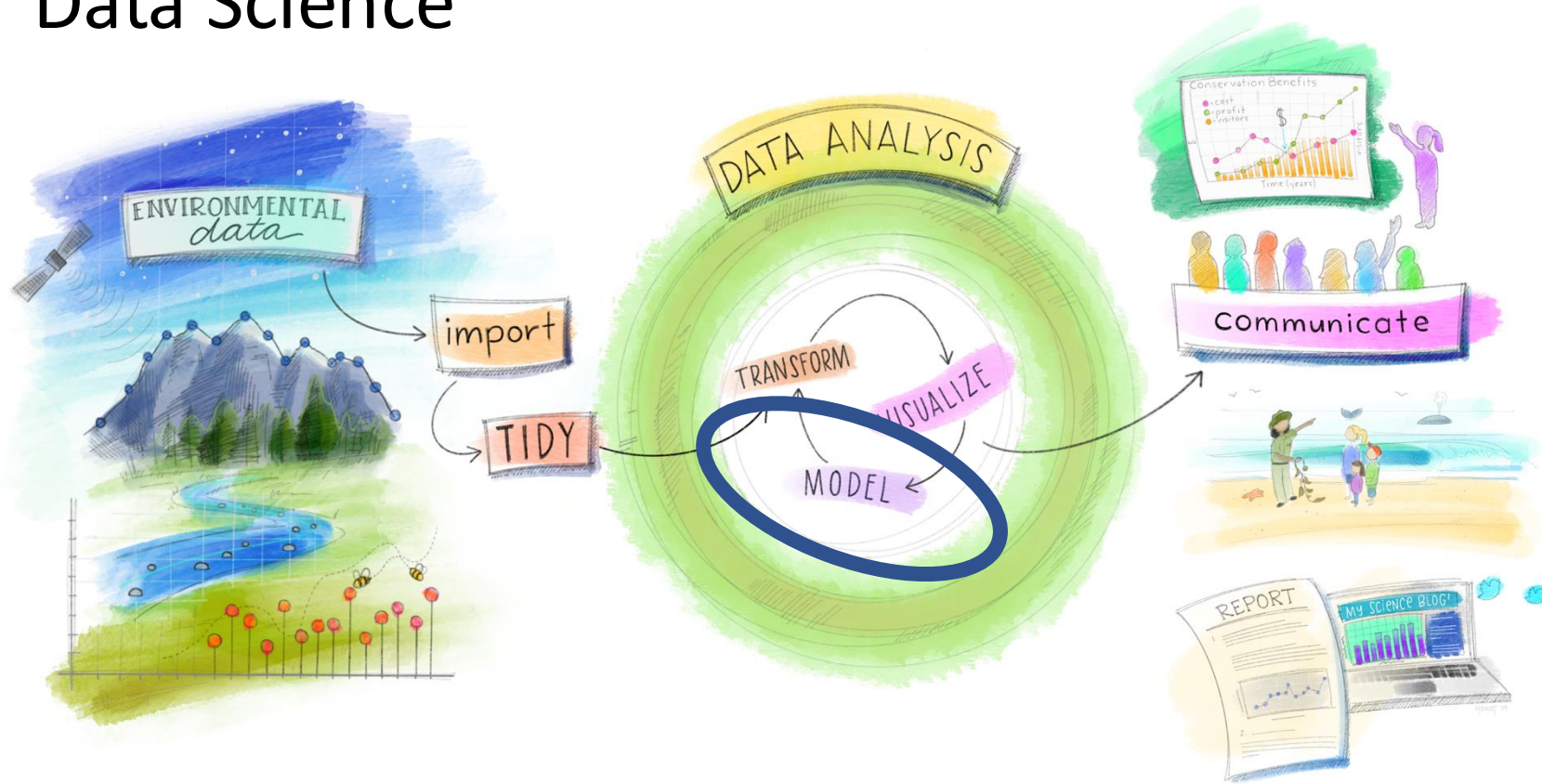
- Free and open source
- Available across platforms
- Huge ecosystem for data manipulation, statistics, visualization, and communication
- Strong research community
- Excellent support for reproducible reports and workflows

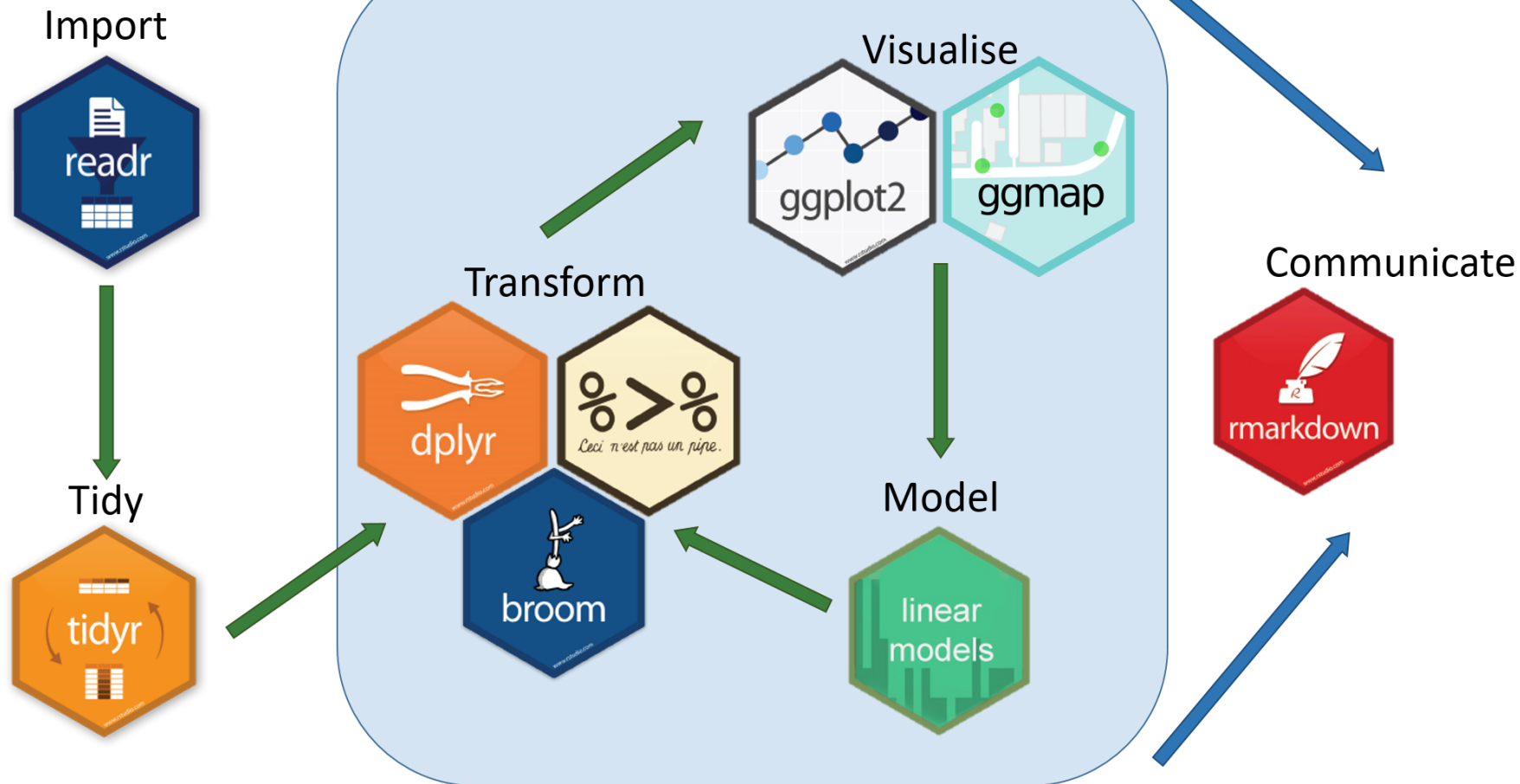
We will focus mainly on the **tidyverse** approach to working with data.

The tidyverse

- An opinionated [collection of R packages](#) designed for data science
- All packages share an underlying design philosophy, grammar, and data structures

Data Science





The course toolkit

Analyze

R + RStudio + tidyverse

Collaborate

Git + GitHub

Communicate

Quarto

And increasingly: **AI-assisted programming**

Example RMarkdowns

- <https://github.com/therkildsen-lab/data-processing>
- <https://github.com/therkildsen-lab/greenland-cod>
- <https://github.com/therkildsen-lab/batch-effect>
- https://github.com/therkildsen-lab/Gal_Sea_cuc

AI in NTRES 6100

Generative AI is changing how people write code and conduct data analysis.

Our goal is **not** to learn how to have AI do the work for us.

We will learn how to use AI while remaining responsible for:

- the scientific question
- analytical choices
- correctness of the code
- interpretation of the result
- reproducibility of the workflow

How lectures and labs differ

Lectures

Build the conceptual and computational foundations.

You will mostly work through problems yourself so you understand **what the code is doing and why**

.

Labs

Apply those foundations to more extended problems.

Labs will be our primary setting for learning to work **productively and critically with AI**

.

Course format

Lectures

Tuesdays and Thursdays, **10:10–11:25 am**

- live coding
- short exercises
- discussion
- assigned readings before class

Labs

Thursdays **or** Fridays, **12:20–2:15 pm**

- hands-on practice
- open-ended problem solving
- AI-assisted programming

Course schedule

- <https://nt246.github.io/NTRES-6100-data-science/syllabus.html>

Live lectures

- Lectures will mostly be live coding, so type along with me!
- Please ask questions! You may ask questions by raising your hand or through the Slack workspace
- This is a safe space and no question is dumb or pointless!

Practice.

Practice.

Practice.

Problem sets

- One problem set each week
- Applies class concepts in a new context
- Due the following **Thursday at 10 pm**
- Submitted through GitHub
- Each assignment will specify whether AI use is **restricted, permitted, or encouraged**

To pass the course

You must:

- participate actively in class
- attend lectures unless other arrangements have been made
- submit at least **7 of 9** problem sets with demonstrated effort
- give a short end-of-course presentation showing how you are applying something from the course in your own work

Communication

Slack

- announcements
- questions
- discussion
- tips and resources

GitHub

- assignments
- repositories
- collaborative coding
- version history

If others may benefit from the answer, ask in a **channel** rather than by direct message.

Asking questions is part of the course

No one arrives with the same background.

- Ask questions during live coding
- Post questions in Slack
- Help answer each other's questions
- Share useful discoveries

You learn a lot by helping someone else solve a problem.

Access and accommodations

Your access to this course is important.

- Students with disability-related accommodations should work with Cornell Student Disability Services and send accommodation letters as early as possible.
- If circumstances affect your ability to participate fully—whether or not you have formal accommodations—please reach out so we can discuss reasonable ways to support your participation and learning.

Zoom participation

This course is designed for **in-person** participation.

- If you are sick or need to be away, we will allow Zoom participation
- But **only if you have made prior arrangements with us**

Ongoing feedback

- We want you to get as much as possible out of this course and there is room for adjustment along the way
 - We always welcome input
 - We will conduct regular quick check-ins
- If anyone has special accessibility concerns, please reach out

Pre-course survey

Please take 5 mins to complete the survey

Link available on the course website under
Resources -> Installation Guide

You are all welcome here

We are committed to a learning environment that is welcoming and supportive regardless of:

- background
- identity
- discipline
- prior programming experience

Everyone will begin this course from a different place. That is expected.

Where to find things

- Course website: <https://nt246.github.io/NTRES-6100-data-science/index.html>
- Canvas page: <https://canvas.cornell.edu/courses/88419>
- Slack: https://join.slack.com/t/ntres-6100-fa26/shared_invite/zt-47thw4thu-GgBchbp_5OnqMZ6T_RGmlA

Check list

Have you:

- ☐ Gotten the **current versions** of R and RStudio working?
- ☐ Followed the instructions for installing Git and making a GitHub account?
- ☐ Joined the workspace on Slack (check that you have all the channels)
- ☐ Added a photo to your GitHub and Slack accounts?
(optional, but encouraged)

Extra office hours this week to help with installation

- Wednesday August 26
 - 11-11.30am (Nina, 208 Fernow Fall)

Introduction to R/RStudio

- RStudio IDE orientation
- Shortcuts and autocomplete
- Install and load packages
- Scripts
- Home directory and RStudio projects
- Change settings to not save workspace